delete or search an element is `O(n)`. This is, however, a very unlikely event that occurs when the hash functions used are properly designed.

The good news is that, if the set of possible keys is static and known in advance, then it is possible to use perfect hashing, and have a worst case O(1) time for all operations.

C.3 Comparative Analysis of Core Data Structures

Now that we have described all the core data structures, we'll try to sum up their characteristics by listing their properties and performance.

The properties we will include are:

- **Order:** Whether a deterministic order for the elements can be maintained. It could be a natural order for the elements, or the insertion order.
- **Unique:** If duplicate elements/keys are forbidden.
- **Associative:** If elements can be indexed by a key
- **Dynamic:** Whether the container can resize on insertion/removal, or its max size needs to be decided in advance.
- **Locality:** Locality of reference: if elements are all stored in a single, uninterrupted block of memory.

Table 2.1 Comparative analysis of core data structures

Structure	Order	Unique	Associative	Dynamic	Locality
Array	yes	no	no	no [§]	yes
Singly-Linked List	yes	no	no	yes	no
Doubly-	yes	no	no	yes	no